

What is CodePipeline..??

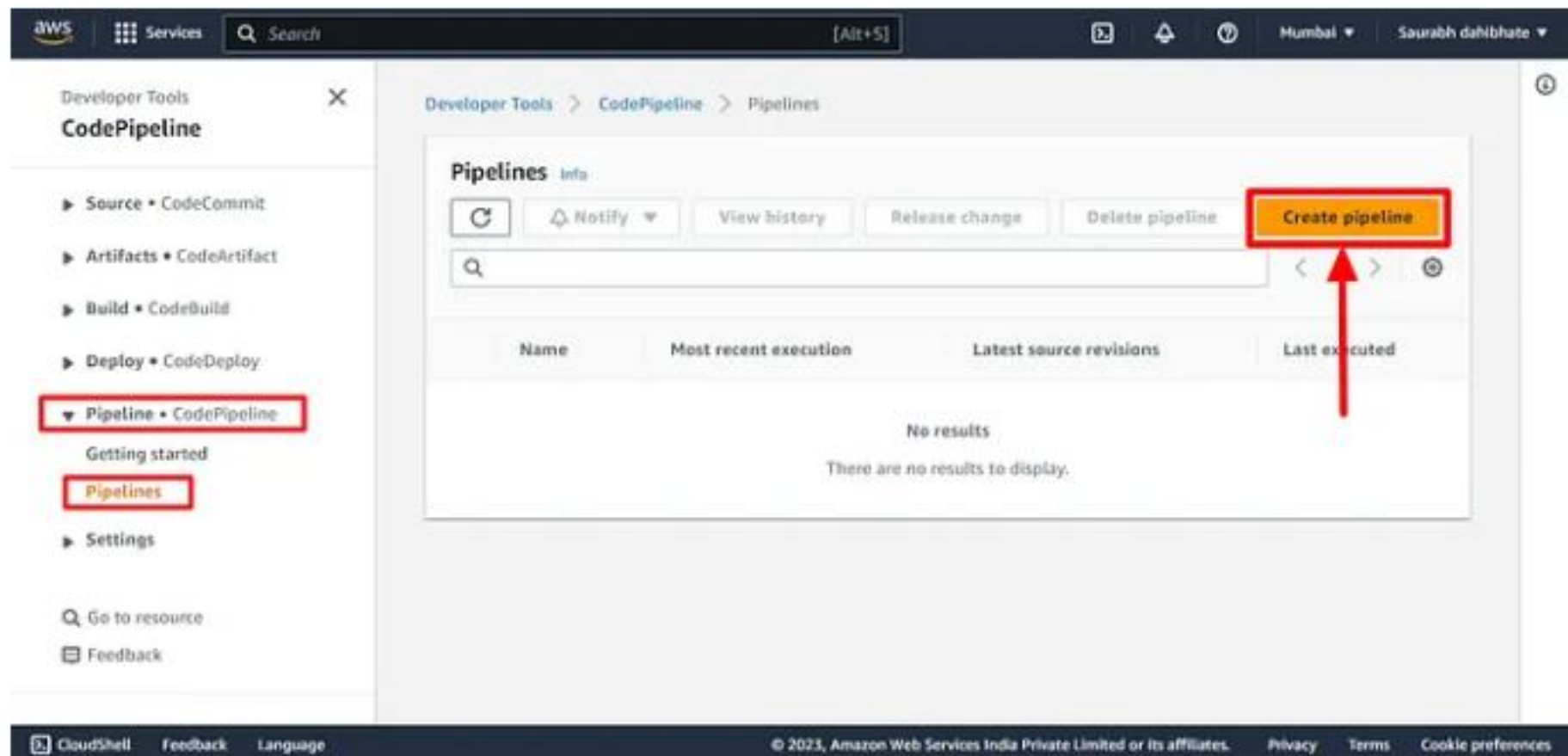
AWS CodePipeline is a continuous delivery service provided by Amazon Web Services (AWS) that helps you automate the software release process for your applications.

With AWS CodePipeline, you can model, visualize, and automate the different stages of your software delivery process, from building and testing to deploying and monitoring.

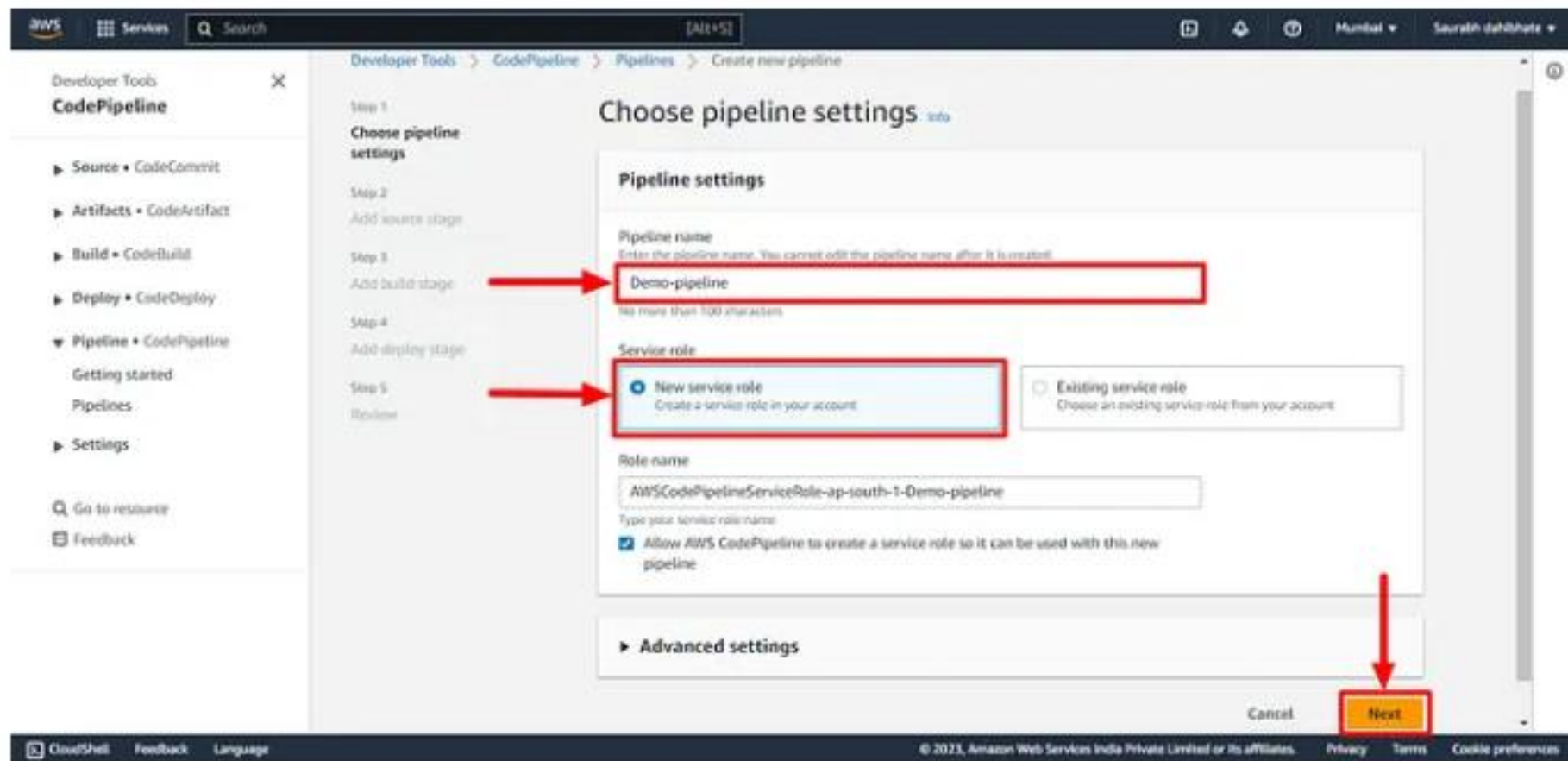
It enables you to rapidly and reliably deliver features and updates to your customers, while reducing the risk of errors and downtime.

- Create a CodePipeline that gets the code from CodeCommit, Builds the code using CodeBuild and deploys it to a Deployment Group.

- This is the last part-04 of CI/CD pipeline. Please refer part 1, 2, 3 of CI/CD for better understanding.
- I have explain the initial stages CodeCommit, CodeBuild, CodeDeploy of pipeline in previous parts.
- Let's start today's task navigate to the CodePipeline section in the AWS management console.



- Provide the Pipeline naming details and let the service role to be default.



- Provide the Code repository details. In our case, we have our code in webapp folder in CodeCommit repository. Choose AWS CodePipeline to automatically execute the pipeline in case of any changes in the code automatically.

Developer Tools

CodePipeline

- Source • CodeCommit
- Artifacts • CodeArtifact
- Build • CodeBuild
- Deploy • CodeDeploy
- Pipeline • CodePipeline
 - Getting started
 - Pipelines
 - Settings

Go to resource

Feedback

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose pipeline settings

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Add source stage

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

AWS CodeCommit

Repository name
Choose a repository that you have already created where you have pushed your source code.

Q Demo-Repository

Branch name
Choose a branch of the repository.

Q main

Change detection options
Choose a detection mode to automatically start your pipeline when a change occurs in the source code.

☐ Amazon CloudWatch Events (recommended)
Use Amazon CloudWatch Events to automatically start my pipeline when a change occurs.

☒ **AWS CodePipeline**
Use AWS CodePipeline to check periodically for changes.

Output artifact format
Choose the output artifact format.

☒ **CodePipeline default**
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include git metadata about the repository.

☐ Full clone
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full git clone. Not supported for AWS CodeBuild actions.

Cancel Previous **Next**

- Now add the Build stage details. Select the build provider and give the project name.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1: Choose pipeline settings

Step 2: Add source stage

Step 3: **Add build stage**

Step 4: Add deploy stage

Step 5: Review

Add build stage Info

Build - optional

Build provider
This is the tool of your build project. Provide build artifact details like operating system, build spec file, and output file names.

Region

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

Environment variables - optional
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#)

Build type

Single build
Triggers a single build.

Batch build
Triggers multiple builds as a single execution.

Cancel Previous Skip build stage **Next**

- Now add the deploy stage. Select the provider and name of the already created application and deployment group.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1: Choose pipeline settings

Step 2: Add source stage

Step 3: Add build stage

Step 4: **Add deploy stage**

Step 5: Review

Add deploy stage Info

Deploy - optional

Deploy provider
Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

Region

Application name
Choose an application that you have already created in the AWS CodeDeploy console. Or create an application in the AWS CodeDeploy console and then return to this task.

Deployment group
Choose a deployment group that you have already created in the AWS CodeDeploy console. Or create a deployment group in the AWS CodeDeploy console and then return to this task.

Cancel Previous Skip deploy stage **Next**

- Now verify the details and create the code pipeline.

Developer Tools

CodePipeline

▶ Source • CodeCommit

▶ Artifacts • CodeArtifact

▶ Build • CodeBuild

▶ Deploy • CodeDeploy

▼ Pipeline • CodePipeline

Getting started

Pipelines

▶ Settings

Go to resource

Feedback

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1

Choose pipeline settings

Step 2

Add source stage

Step 3

Add build stage

Step 4

Add deploy stage

Step 5

Review

Review

info

Step 1: Choose pipeline settings

Pipeline settings

Pipeline name

Demo-pipeline

Artifact location

A new Amazon S3 bucket will be created as the default artifact store for your pipeline.

Service role name

AWSCodePipelineServiceRole-ap-south-1-Demo-pipeline

Step 2: Add source stage

Source action provider

Source action provider

AWS CodeCommit

RepositoryName

Demo-Repository

BranchName

main

PollForSourceChanges

true

OutputArtifactFormat

CODE_ZIP

Step 3: Add build stage

Build action provider

Build action provider

AWS CodeBuild

ProjectName

Demo-app

Step 4: Add deploy stage

Deploy action provider

Deploy action provider

AWS CodeDeploy

ApplicationName

Demo-app

DeploymentGroupName

Demo-app-group

Cancel

Previous

Create pipeline

- The pipeline will now fetch the code from the CodeCommit repository.
- Then, it will build the code and deploy in the server as we can see in the below screenshot.

The screenshot displays the AWS CodePipeline console interface. At the top, a green banner indicates 'Success: Congratulations! The pipeline Demo-pipeline has been created.' Below this, the 'Demo-pipeline' is shown with three stages: 'Source', 'Build', and 'Deploy'. Each stage is marked as 'Succeeded' with a green checkmark. Red arrows point to the 'Source', 'Build', and 'Deploy' stage names. The 'Source' stage uses 'AWS CodeCommit' and is 'Succeeded - 11 minutes ago'. The 'Build' stage uses 'AWS CodeBuild' and is 'Succeeded - 9 minutes ago'. The 'Deploy' stage uses 'AWS CodeDeploy' and is 'Succeeded - 7 minutes ago'. The left sidebar shows the 'CodePipeline' service selected, with 'Pipeline' highlighted under the 'Pipeline' section. The bottom of the console shows the footer with 'CloudShell', 'Feedback', 'Language', and copyright information for Amazon Web Services India Private Limited.

- Now, navigate to the public URL of the instance and check the Webpage.

Welcome to my website

This is a simple website hosted on AWS CodeCommit.

- We have successfully built the pipeline. Let's witness the magic. Let's modify our code and view the pipeline.
- I have changed my index.html file and committed it in the CodeCommit repository.


```
File Edit Selection View Go Run Terminal Help
index.html - Untitled (Workspace) - Visual Studio Code

EXPLORER
  UNFILED (WORKSPACE)
  Demo-Repository
    assets
    scripts
    appspec.yml
    buildspec.yml
    index.html

index.html
1 <!DOCTYPE html>
2 <!--[if IE 10]><html class="ie ie10" lang="en"><![endif]-->
3 <!--[if IE 7]><html class="ie ie7" lang="en"><![endif]-->
4 <!--[if IE 8]><html class="ie ie8" lang="en"><![endif]-->
5 <!--[if (gte IE 9)!(IE)]><!-->
6 <html lang="en">
7 <!--<![endif]-->
8 <head>
9   <meta charset="utf-8">
10  <meta name="viewport" content="width=device-width, initial-scale=1, maximum-scale=1">
11  <meta name="description" content="">
12  <meta name="author" content="">

TERMINAL
create mode 100644 assets/img/prettyPhoto/light_square/sprite.png
create mode 100644 assets/img/team/team1.png
create mode 100644 assets/img/team/team2.jpg
create mode 100644 assets/js/bootstrap.min.js
create mode 100644 assets/js/jquery.js
create mode 100644 assets/js/jquery.prettyPhoto.js
create mode 100644 assets/js/scrollReveal.js
create mode 100644 assets/scripts/custom.js
PS C:\Users\saura\Desktop\Dec-proj\Demo-Repository>
PS C:\Users\saura\Desktop\Dec-proj\Demo-Repository> git push origin main
Enumerating objects: 74, done.
Counting objects: 100% (74/74), done.
Delta compression using up to 4 threads
Compressing objects: 100% (71/71), done.
Writing objects: 100% (72/72), 512.90 KiB | 4.58 MiB/s, done.
Total 72 (delta 6), reused 0 (delta 0), pack-reused 0
remote: Validating objects: 100%
To https://git-codecommit.ap-south-1.amazonaws.com/v1/repos/Demo-Repository
   ae70301..1c2b0fa  main -> main
PS C:\Users\saura\Desktop\Dec-proj\Demo-Repository>
```

- The pipeline has automatically picked up the code from the repository, built and deployed it in the server.

The screenshot displays the AWS CodePipeline console for a pipeline named 'Demo-pipeline'. The pipeline is in a 'Succeeded' state, as indicated by the green checkmark and the 'Succeeded' status in the top bar. The pipeline execution ID is '607304c4-6720-4115-b056-06a298a53335'. The pipeline consists of three stages: Source, Build, and Deploy, all of which are marked as 'Succeeded'. Red arrows point to the 'Succeeded' status and the 'Source: new web page' message in each stage. The interface includes a left-hand navigation menu with options like Source, Artifacts, Build, Deploy, and Pipeline. The top bar shows the AWS logo and the user's name. The bottom bar contains the CloudShell logo and copyright information.

Developer Tools > CodePipeline > Pipelines > Demo-pipeline

Demo-pipeline [Notify] [Edit] [Stop execution] [Clone pipeline] [Release change]

Source Succeeded
Pipeline execution ID: 607304c4-6720-4115-b056-06a298a53335

Source
AWS CodeCommit
Succeeded - 3 minutes ago
1c2b40a: Source: new web page

Disable transition

Build Succeeded
Pipeline execution ID: 607304c4-6720-4115-b056-06a298a53335

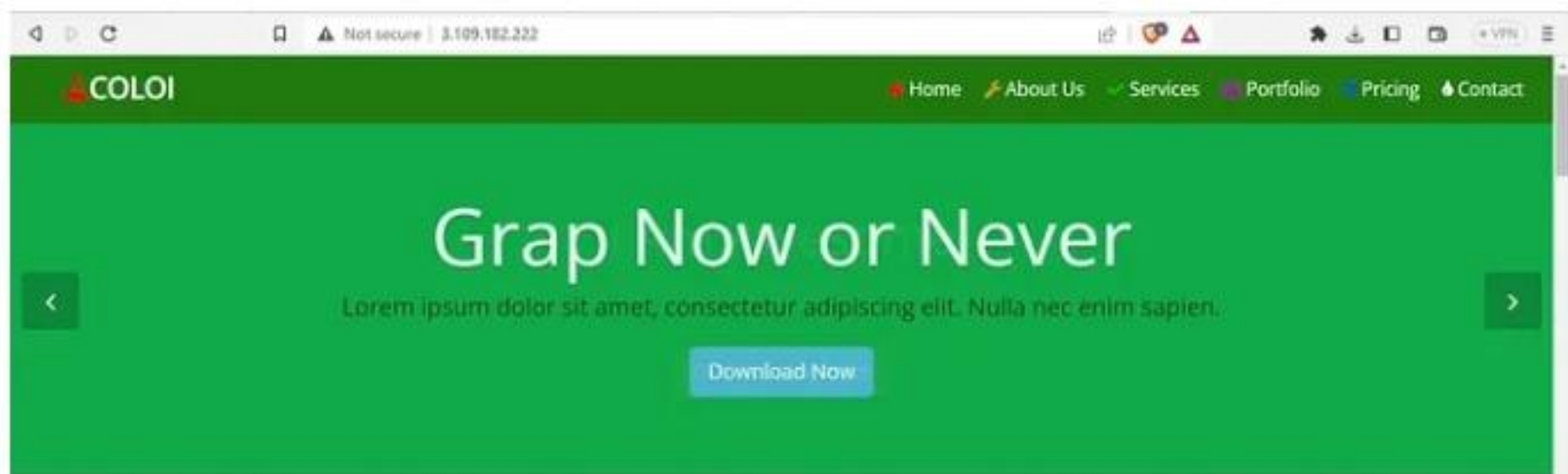
Build
AWS CodeBuild
Succeeded - 1 minute ago
Details
1c2b40a: Source: new web page

Disable transition

Deploy Succeeded
Pipeline execution ID: 607304c4-6720-4115-b056-06a298a53335

Deploy
AWS CodeDeploy
Succeeded - 1 minute ago
Details
1c2b40a: Source: new web page

- Now check the public URL. Yes, I can see my new application with an eye-catching webpage.



Meet Team

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Romiz Nopvok

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Jhon Deo

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Mark Kimza

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Nopen Riox

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Services

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Responsive Design



24x7 Support



Awesome Display

Services

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Responsive Design

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24x7 Support

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Awesome Display

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Tested Template

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Say Yes To Customize

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Well Documented

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Portfolio

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Starter

\$ 100

SGS Storage

1GB Space

200 Bandwidth

100 Email Address

24x7 Support

Live Chat

[Signup](#)

Medium

\$ 200

5GB Storage

1GB Space

200 Bandwidth

100 Email Address

24x7 Support

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\$ 300

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200 Bandwidth

100 Email Address

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